Unit no. º 11

CROSS-TABULATING TWO CATEGORICAL VARIABLES

1-12 March 2027

I Two-way tables and marginal frequencies

Definitions

A qualitative variable describes a category. It is nominal if its categories have no natural order (means of transport), ordinal if they are ordered (level of satisfaction). A two-way table counts individuals according to two variables. The row and column totals are the margins. A marginal frequency is a margin divided by the total count.

Example 11.1

In a population of 100 individuals, two characteristics are observed A, B :

	A	$\overline{A}$	Total
B	20	15	35
$\overline{B}$	12	53	65
Total	32	68	100

Calculate the marginal frequencies of A, B, $\overline{B}$, then the frequency of individuals satisfying both A and B.

Solution:

$f(A) = 32/100 = 0{,}32$; $f(B) = 35/100 = 0{,}35$; $f(\overline{B}) = 65/100 = 0{,}65$. The frequency of $A \cap B$ is $20/100 = 0{,}20$; this is not a marginal frequency.

Example 11.2

A Première class of 32 students is divided as follows. Complete the margins and calculate the marginal frequencies "NSI" and "girls".

	Boys	Girls	Total
NSI	12	9	**21**
No NSI	6	5	**11**
Total	**18**	**14**	32

Solution:

f(NSI) = 21 $/32 = 65{,}625\,\%$; f(girls) = 14 $/32 = 43{,}75\,\%$.

II Conditional frequencies and reading sentences

Definition

The frequency of A **among the individuals of** B, if this subpopulation is not empty, is

$$f_B(A) = \frac{\text{count satisfying } A \text{ and } B}{\text{count of } B}.$$

The denominator corresponds to the population introduced by "among".

Example 11.3

In the NSI table, calculate the frequency of girls among pupils without NSI, then the frequency of pupils without NSI among the girls. Are the two results equal?

Solution:

Of the 11 pupils not taking Computer Science (NSI), 5 are girls: $5/11 \approx 45{,}45\%$. Of the 14 girls, 5 do not take NSI : $5/14 \approx 35{,}71\%$. The numerator is the same; the reference populations differ.

Example 11.4

In the table of 100 individuals, calculate $f_B(A)$, $f_A(B)$ and $f_{\overline{A}}(B)$. Express each result as a sentence.

Solution:

$f_B(A) = 20/35 = 4/7 \approx 57{,}14\%$ of the individuals in B satisfy A. $f_A(B) = 20/32 = 62{,}5\,\%$ of individuals of A satisfy B. $f_{\overline{A}}(B) = 15/68 \approx 22{,}06\,\%$ of individuals not satisfying A satisfy B.

Example 11.5

Does "60% of cyclists are students" allow us to conclude that 60% of students are cyclists?

Solution:

No. For example, out of 10 cyclists, 6 are pupils; if the population has 100 pupils in total, only 6% of the pupils are cyclists.

III Complete a table from frequencies

Example 11.6: Fictional situation

A group has 200 people, 120 of whom use a bus (B). Of these 120 people, 75% own a subscription (A). Of the other 80, 25% have a subscription.

1. Construct the two-way table $B/\overline{B}$ and $A/\overline{A}$ with its margins.

2. Calculate the marginal frequency of A.

3. Calculate the frequency of B among A.

Solution:

In B, $120 \times 0{,}75 = 90$ people satisfy A, so 30 do not satisfy it. In $\overline{B}$, $80 \times 0{,}25 = 20$ satisfy A, and 60 do not satisfy it.

	A	$\overline{A}$	Total
B	90	30	120
$\overline{B}$	20	60	80
Total	110	90	200

$f(A) = 110/200 = 55\,\%$. $f_A(B) = 90/110 = 9/11 \approx 81{,}82\,\%$.

Example 11.7

A headline states "Three out of four subscribers take the bus", based on $f_B(A) = 75\ \%$. Explain the error and suggest a correct sentence.

Solution:

The headline reverses the conditioning. The correct sentence is: "Three out of four people among those who take the bus have a season ticket." The proportion of season-ticket holders who take the bus is here approximately $81{,}82\,\%$.

IV Process a file and collect real data

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# Exemple fictif : (transport, satisfaction)
donnees = [("bus", "oui"), ("marche", "oui"),
           ("bus", "non"), ("velo", "oui"),
           ("bus", "oui"), ("marche", "non")]

bus = [r for r in donnees if r[0] == "bus"]
bus_satisfaits = [r for r in donnees
                  if r[0] == "bus" and r[1] == "oui"]
autres = [r for r in donnees if not r[0] == "bus"]
```

Example 11.8

Give the counts of the three lists. Calculate the satisfaction frequency among bus users.
Replace `and` by `or` in the second filter: how many rows are then selected?

Solution:

Counts: 3 , 2, 3. Among bus users, the satisfaction frequency is 2 /3. With `or`, we select the 3 bus users and the 2 satisfied people using another means: 5 rows, without double-counting the rows that satisfy both criteria.

Activity with class data

Anonymously collect two variables: main means of transport and journey time by ordered categories ($<$ 15 min; 15 to under 30 min; $\geq$ 30 min). One row per response, without name. Build the two-way table, then two conditional frequencies in opposite directions. Present sample size, date of collection and any non-responses.

Example 11.9

Carry out the data collection and write two justified conclusions. Explain why the results do not automatically describe all students in Madagascar.

Solution:

The values depend on the actual data collection. Check: one category per variable and per answer; sum of cells equal to the number of usable responses; correct denominator for each frequency; rounding stated. The sample is one particular class, not a representative sample of all the pupils in the country. Non-responses must be distinguished from an observed category.